Chen Sun

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EDUCATION

University of Southern California **Los Angeles, CA**
Ph.D. student, Biomedical Engineering Sept 2023-April 2028

Advisor: Dr. Dong Song

Queens's University **Ontario, Canada**
B.S. degree: Computer Science Sept 2018-May 2022

SKILLS

Technical Skills: Python · PyTorch · MATLAB · SQL · ROS · Go · OpenCV
Expertise: Time-Series Analysis · Signal Processing · Spiking Neural Networks · Nonlinear & Nonstationary Modeling · Reinforcement Learning · Deep Learning

ACADEMIC EXPERIENCE

Neural Modeling and Interface Laboratory, USC **Los Angeles, CA**
Research Assistant Sept 2023-Present

- Led DARPA INSPIRE project: Identify synaptic plasticity learning rules from endogenous time-series recordings in behaving animals.
  - Built PyTorch model to directly infer population-level LTP/LTD from hippocampal time-series recorded during cognitive learning tasks
  - Built analysis pipeline for preprocessing and processing raw neural data

Department of Biomedical Engineering, USC **Los Angeles, CA**
Teaching Assistant Sept 2023-Present

- BME 501-Intro to Biomedical Engineering, BME 513-Signal and System Analysis, BME 552-Neural Implant Engineering

Machine Intelligence & Biocomputing Laboratory (MIB), Queen's University **Ontario, Canada**
Research Assistant March 2022-April 2023

- Investigated the application of various evolutionary algorithms for pruning deep spiking neural networks (DSNNs).
- Researched the use of coevolution algorithms for optimizing DSNNs with multi-objective models.
- Examined the integration of dendrites into SNN models and developed a continuous learning capable dendritic DSNN model.

PROFESSIONAL EXPERIENCE

ByteDance **Beijing, China**
Full-Stack Developer Intern May 2021-Sept 2021

Developed backend systems for two internal platforms:

- Form-Filling Platform: Designed with Python's Django framework and Gunicorn for concurrency, enabling secure data collection from isolated informants via Lark app and online Excel integration, supporting over 100 active users.
- HR Management Platform (HR+): Built with Golang's Hertz framework, including an admin module for internal operations and an internal transfer hunting platform for HR use. The platform was deployed company-wide, serving HR operations across all departments.

Mesoor AI **Shanghai, China**
Web Crawler Developer Intern Aug 2020-Feb 2021

- Designed and developed a modular recruitment recommendation system to automate resume collection and job matching, while implementing anti-crawling strategies to bypass website restrictions.
- Improved a BERT NLP model for resume analysis and created targeted web crawlers to gather market data, enhancing the platform's accuracy and efficiency.
- Built a post-promotion service and secure proxy generators for data requests, ensuring system scalability and user engagement.

SELECTED PROJECTS

LitAgent: Agentic Literature Intelligence Platform **Los Angeles, CA**
Personal Project Apr 2026-Present

- Built LitAgent , an end-to-end RAG-based literature assistant that ingests Zotero libraries, indexes papers, and delivers grounded QA with citations using user-provided LLMs as agents.
- Developed interactive wiki and relation-map interfaces with retrieval, graph construction, and orchestration modules for cross-paper synthesis.

GazeReader: Apple Vision Pro PDF Reader App **Los Angeles, CA**
Personal Project Nov 2025-Dec 2025

- Developed GazeReader, immersive ARVR application on visionOS using Swift, RealityKit, and ARKit.
- Implemented spatial PDF viewer with real-time highlighting of read text sections via eye-tracking and hand gestures.

EEG Foundation Challenge **Los Angeles, CA**
Deep Decoding Model for Cross-Task Cross-Subject EEG Decoding Sept 2025-Dec 2025

- Developed PyTorch model to predict behavioral performance metrics from EEG data.
- Incorporated subject- and session-specific embeddings to model individual variability.
- Designed neural masking mechanism: session embedding passed through 2-layer MLP to generate per-session neural mask, aggregating subject-specific channels into a composite neural signal representing overall EEG features.
- Ranked top 10th percentile among all participating teams.

Massachusetts Institute of Technology **Cambridge, MA**
Thalamic Model for Task Representation and Segmentation in Dynamic Environments June 2023-Present

- Built PyTorch thalamic model for task representation and segmentation in dynamic environments with self-generated internal representations
- Trained in custom OpenAI Gym setup simulating volatile conditions and task switching to study task boundary inference triggers
- Showed single unsegmented network could learn opposing tasks, achieve lifelong learning, and adapt to novel tasks without catastrophic forgetting
- Generated predictions on task segmentation triggers in animal behavior; compared synaptic plasticity to neural activity updates

MIB Lab **Ontario, Canada**
Cooperative Coevolution Pruning for Spiking Neural Networks June 2022-March 2023

- Conducted research on pruning SNNs using a Cooperative Coevolution algorithm (CCEP).
- Constructed a two-layer SNN using Spikingjelly and trained it using surrogate gradient descent.
- Modified original CCEP code for spiking neural network pruning.
- Compressed the model through pruning by treating each Conv2D layer as a subpopulation and applying cooperative evolution.

Queens's University **Ontario, Canada**
EvoMedley, Interactive Melody Generator with Multi-song Fusion Jan 2022-April 2022

- Proposed EvoMedley, an interactive music mixing method using evolutionary algorithms (EA) to generate personalized music from user-selected playlists.
- Developed a feature extraction algorithm to decompose MIDI files into minimal melodic fragments, creating a gene pool for EA-based music generation.
- Designed a dynamic fitness function guided by user feedback, enabling real-time adaptation to user preferences while preserving the characteristics of the original tracks.

Tangent **Ontario, Canada**
Customized Online Archive *May 2022-Sept 2022*

- Led online archive project as part of tangent Collective art organization, managing recruitment, project management, backend development, and service deployment.
- Implemented backend using NGINX, WSGI, Python Django, and MySQL.
- Deployed service on AWS EC2 server with AWS S3 data storage, CloudFront CDN, and AWS RDS database for data recoverability. Ensured stability through pm2 automatic deployment and daemonization.